

Assignment 2: Network Robustness

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1 Introduction

This assignment investigates the robustness of different networks under progressive node removal attacks. Robustness is evaluated by monitoring the size of the giant component (GC) during node removal. A network is considered robust if the GC remains large even after many nodes are removed.

Three networks are analyzed:

- an Erdős–Rényi random graph (ER);
- a Barabási–Albert scale-free graph (BA);
- a real-world anime similarity network built from the MyAnimeList dataset.

Four attack strategies are applied:

- random node removal;
- highest degree node removal;
- highest PageRank node removal;
- highest betweenness node removal (approximate).

Finally, three robustness-enhancing strategies are applied to the anime network, and their impact is evaluated through the critical threshold f_c and the collapse dynamics.

2 Attack Methodology

Given a graph G , the attack procedure is iterative:

1. compute the largest connected component (GC);
2. select a batch of nodes according to the attack strategy;
3. remove the selected nodes;
4. recompute the GC.

The robustness measure is:

$$\frac{G}{N},$$

where G is the size of the GC after removal and N is the initial number of nodes. A slow decrease of G/N indicates robustness, while a rapid decrease indicates vulnerability.

The collapse point is defined as the moment when the GC falls below 50% of its original size.

3 Attack Strategies

1. **Random attack:** nodes are removed uniformly at random.
2. **Degree attack:** nodes with highest degree are removed first.
3. **PageRank attack:** nodes with highest PageRank are removed first.
4. **Betweenness attack:** nodes with highest betweenness centrality are removed first.

Due to computational cost, an approximate version is used.

4 Experiments on Synthetic Networks

4.1 Erdős–Rényi Network

The ER graph ($N = 1000$, $p = 0.01$) shows homogeneous robustness. Random failures cause a gradual decrease of the GC, while targeted attacks produce a faster collapse. This behavior is expected because ER networks lack dominant hubs.

Collapse points:

- Random: 500 nodes removed,
- Degree: 500 nodes removed,
- PageRank: 500 nodes removed,
- Betweenness: 500 nodes removed.

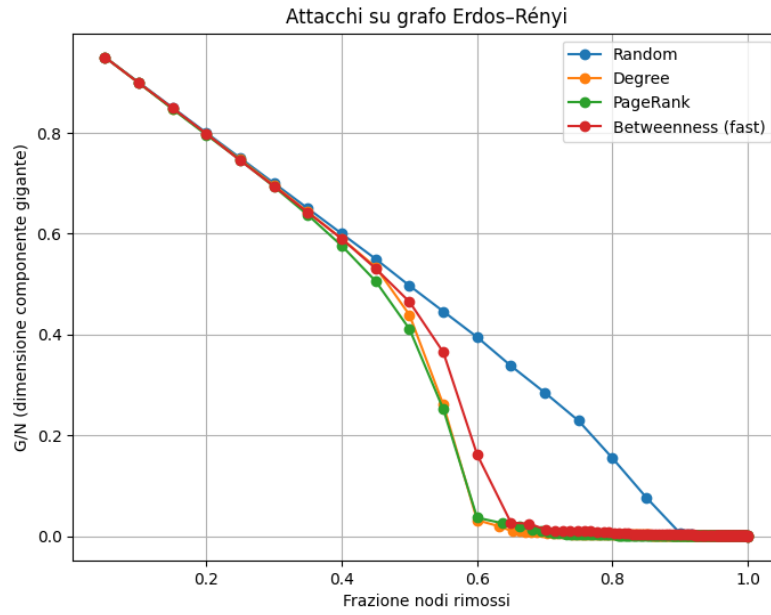


Figure 1: Attack results on the Erdős–Rényi graph.

4.2 Barabási–Albert Network

The BA graph ($N = 1000$, $m = 5$) behaves differently. Random failures have limited impact, while targeted attacks rapidly fragment the network. This confirms the vulnerability of scale-free networks to hub removal.

Collapse points:

- Random: 500 nodes removed,
- Degree: 325 nodes removed,
- PageRank: 325 nodes removed,
- Betweenness: 350 nodes removed.

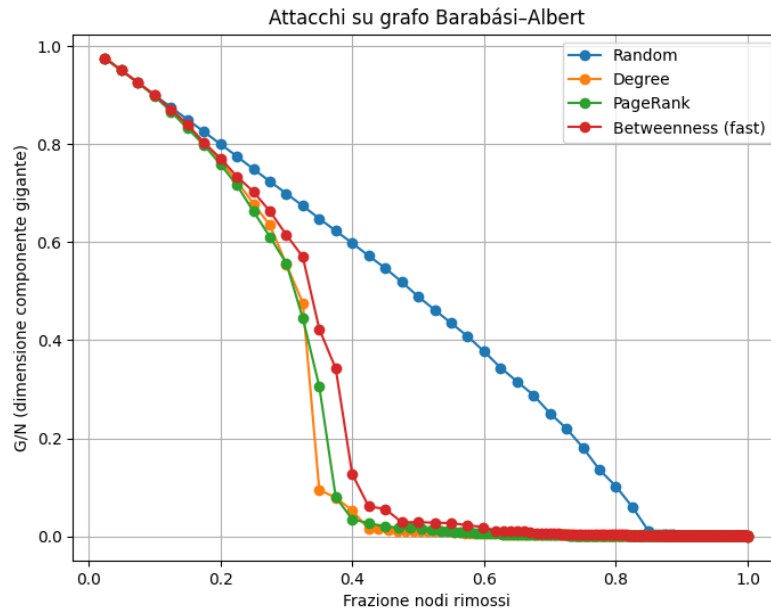


Figure 2: Attack results on the Barabási–Albert graph.

5 Anime Similarity Network

The anime network contains:

- 3579 nodes,
- 780393 edges.

Edges connect anime sharing at least two genres. The network is extremely dense and highly clustered.

Collapse points:

- Random: 1200 nodes removed,
- Degree: 1200 nodes removed,
- PageRank: 1200 nodes removed,
- Betweenness: 1000 nodes removed.

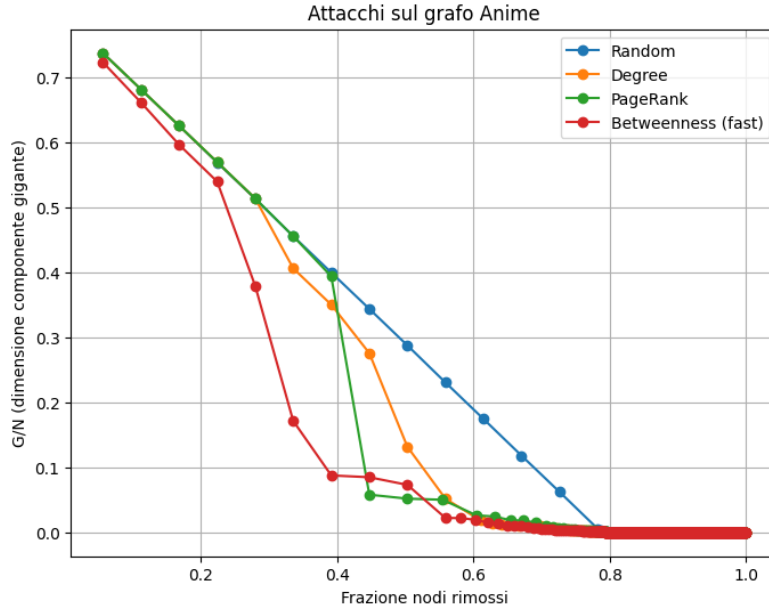


Figure 3: Attack results on the anime similarity network.

The anime network is highly robust: even targeted attacks require removing about one third of the nodes before collapse. Betweenness is the most effective strategy because it identifies nodes acting as bridges between communities.

6 Building Robustness

Three strategies are applied to improve robustness:

1. **Ring Degree:** add edges among neighbors of the top-degree nodes.
2. **Ring Betweenness:** add edges among neighbors of the top-betweenness nodes.
3. **High Clustering:** add edges between nodes with highest clustering coefficient.

6.1 Critical Threshold

The critical threshold is:

$$f_c = 1 - \frac{1}{\frac{k_2}{k_1} - 1}.$$

The results obtained are:

Graph	N	L	k_1	k_2	f_c
Original	3579	780393	436.10	3.52×10^5	0.998758
Ring Degree (top-3)	3579	1720329	961.35	1.66×10^6	0.999419
Ring Betweenness (top-3)	3579	2516321	1406.16	3.07×10^6	0.999542
High Clustering (200 edges)	3579	780593	436.21	3.52×10^5	0.998758

Table 1: Critical threshold comparison after robustness strategies.

Ring Degree and Ring Betweenness significantly increase k_1 and k_2 , producing higher f_c . High Clustering has negligible effect.

Ring Betweenness is the strategy that produces the largest theoretical improvement because it increases redundancy around the nodes that act as bridges between communities. These nodes have the highest betweenness centrality and are structurally critical for global connectivity. By adding edges among their neighbors, the network gains alternative paths in its most fragile regions, which significantly increases the second moment k_2 and therefore the critical threshold f_c . This makes the network more robust against random failures than any other strategy.

6.2 Collapse Dynamics After Robustness Strategies

After applying robustness strategies, attacks are repeated. The collapse points are:

- Original graph: 1200 nodes removed (Degree attack),
- Ring Degree: 800 nodes removed (Degree attack),
- Ring Betweenness: 1000 nodes removed (Degree attack).

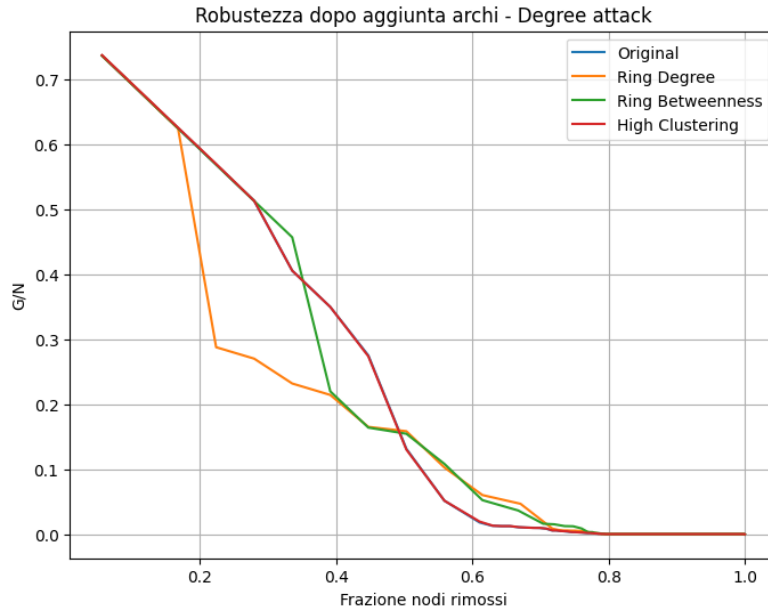


Figure 4: Comparison between original and improved networks under attacks.

Adding edges around hubs (Ring Degree) makes those hubs even more structurally critical, causing earlier collapse under degree-based attacks.

High Clustering does not significantly change collapse behavior.

7 Discussion

The experiments confirm theoretical expectations:

- ER networks show homogeneous robustness;
- BA networks are robust to random failures but vulnerable to targeted attacks;
- the anime network is extremely robust due to high density and clustering.

Robustness-enhancing strategies behave differently:

- Ring Degree and Ring Betweenness increase theoretical robustness;
- Ring Degree becomes more vulnerable to degree attacks;
- High Clustering has negligible effect.

This highlights the importance of edge placement: improving redundancy around hubs may increase vulnerability to targeted attacks.

8 Conclusion

The analysis carried out in this assignment highlights how network robustness strongly depends on the structural properties of the underlying graph and on the nature of the attack strategy.

Random failures are generally less harmful than targeted attacks: when nodes are removed uniformly at random, all networks preserve a large portion of their giant component for a long time. In contrast, attacks based on degree, PageRank or betweenness centrality rapidly fragment the network by removing structurally important nodes.

The Barabási–Albert graph exhibits the typical behaviour of scale-free networks: it is highly resilient to random failures but collapses quickly when hubs are removed. The Erdős–Rényi graph shows homogeneous robustness, with all attack strategies producing similar collapse points due to the absence of dominant nodes.

The anime similarity network demonstrates exceptional robustness. Its extremely high density and clustering coefficient create a large number of redundant paths, allowing the network to withstand even aggressive targeted attacks. Only betweenness-based removal produces a noticeably earlier collapse, confirming the structural importance of bridge nodes connecting different communities.

Robustness-enhancing strategies have different effects. Ring Degree and Ring Betweenness increase the theoretical robustness by raising the critical threshold f_c , but they do so in different ways. Ring Betweenness provides the largest improvement because it adds redundancy around nodes that act as bridges between communities, strengthening the most fragile regions of the network. Conversely, Ring Degree may unintentionally increase vulnerability to degree-based attacks by making hubs even more central. High Clustering has negligible impact, as it modifies only local neighbourhoods without affecting global connectivity.

Overall, the experiments show that robustness is not determined solely by network size or density, but by how connections are organized and how critical nodes contribute to global connectivity. Understanding these structural mechanisms is essential when designing networks that must remain functional under failures or targeted attacks.